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Lesson 2: Data Exploration and Preparation

Technologies for Artificial Intelligence

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Before we start

Exercise 1 was set last week. We discuss it now.

- The wine-quality workflow, end to end
- What counts is **methodology**, not accuracy
- A show of hands: did you predict a class, or a number?

What today builds on

- Lesson 1’s rule: **nothing is learned before the split**
- Lesson 1’s tool: `Pipeline`, which enforces that rule in the code instead of relying on memory
- Today: both, on messy data - gaps, extreme values, text categories

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Today

- Exploratory analysis
- Missing values
- Outliers
- Scaling and encoding
- Feature engineering and pipelines
- Leakage that raises no error and looks like a good result

One dataset, all lesson long

2000 synthetic telecom customers. Will they **churn** (cancel their contract)?

- Numeric: tenure, monthly charges, age, support calls
- Categorical: contract type, region, zip code (493 different values)
- Missing values, extreme values, and one column built to have **no relation** to churn

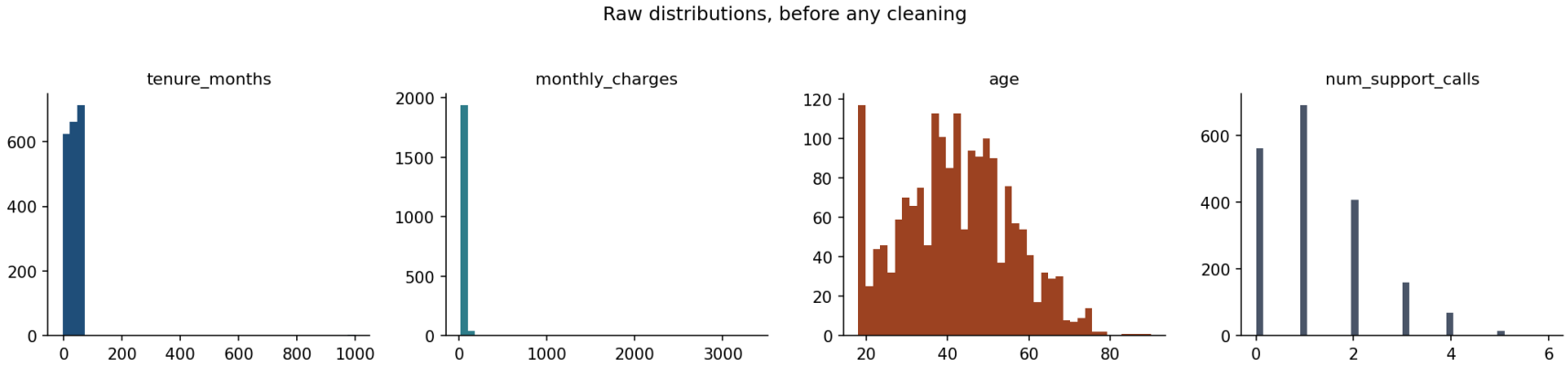
Look before you touch anything

First, always: `.info()`, `.describe()`, and the share of each class.

- 2000 rows, 8 columns, numbers and text
- Churn rate **19.4%**: always predicting “no churn” is right 80.6% of the time
- `tenure_months` reaches 999 months, 83 years: an error in the data

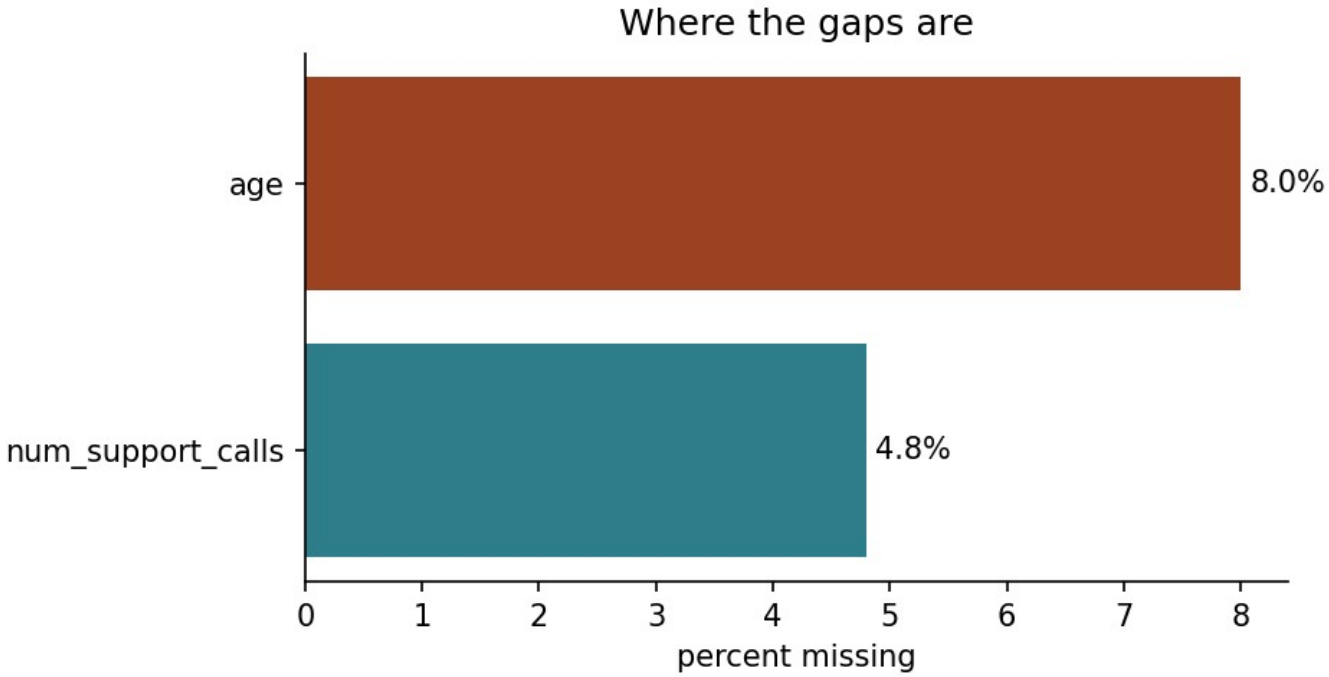
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Outliers squash two of these four



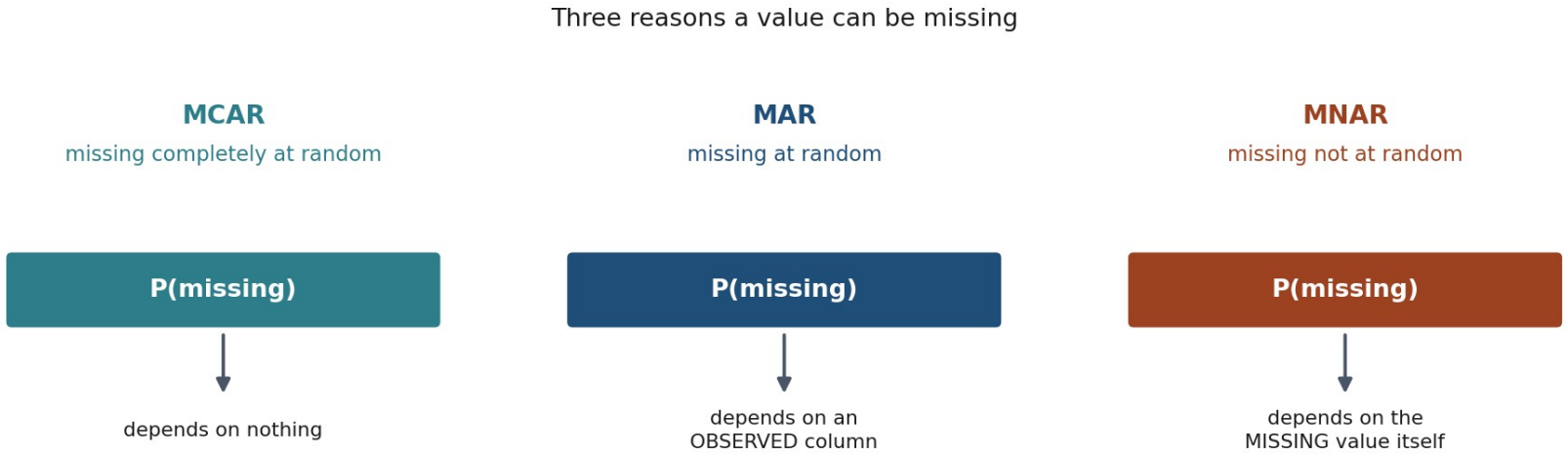
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Where the gaps are



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Three reasons a value is missing



age: missing completely at random

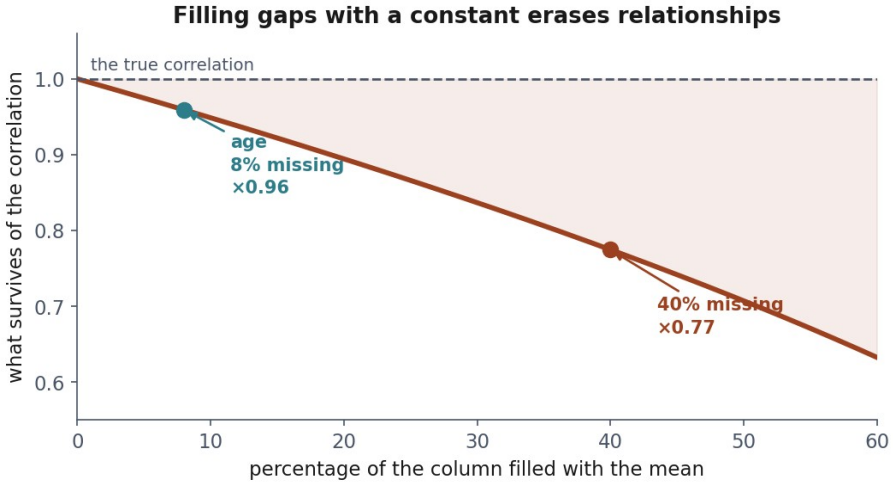
MCAR, for short: some customers left it blank, for no reason related to any column.

- Filling the gaps with the mean leaves the mean of `age` unbiased
- So is filling free? The next slide shows what it costs

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What mean imputation actually costs

Filling with the mean weakens every correlation `age` has with other columns, even under MCAR.



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Support calls: missing at random

MAR, for short: long-tenure customers leave it blank more often.

- Dropping those rows would remove mostly long-tenure customers
- A **missingness indicator** (1 if missing, 0 if not) adds nothing here: `tenure_months` already tells the model which rows tend to have a gap

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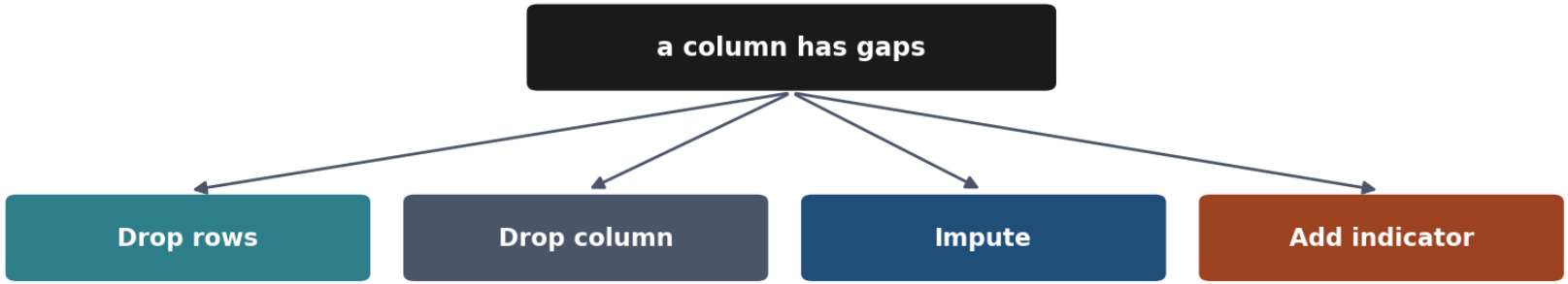
What to do about a gap

- **Drop rows**: only if few, and only if MCAR
- **Drop the column**: if it is missing too often to help
- **Impute**: mean/median or most-frequent with `SimpleImputer`, neighbours with `KNNImputer`
- **Add a missingness indicator**: only if *being missing* tells the model something the other columns cannot - typically when missing not at random (MNAR)

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Choosing among the four

Which one you pick is a question about why the data is missing



the mechanism decides, and every option except dropping the column learns a statistic, so it is fitted on the training data only

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Extreme does not mean wrong

- `tenure_months` runs **-3 to 999**; ordinary customers sit at 0 to 72
- Three reasons: a **recording error**, a **rare but genuine** value, a **different population**
- A monthly charge of 3,344.7 may be a typo or a corporate account: the numbers alone cannot tell
- Two rules follow, one on standard deviations, one on quartiles: both only **flag** a value for a person to check

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Rule 1: distance in standard deviations, flag beyond k = 3

$$z_i = \frac{x_i - \bar{x}}{s}$$

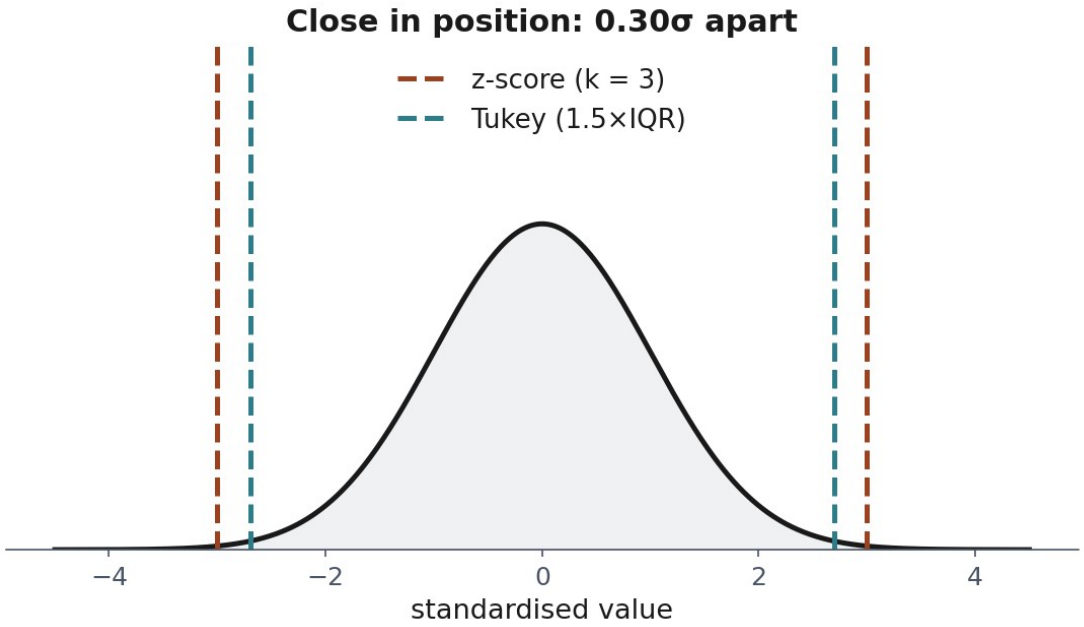
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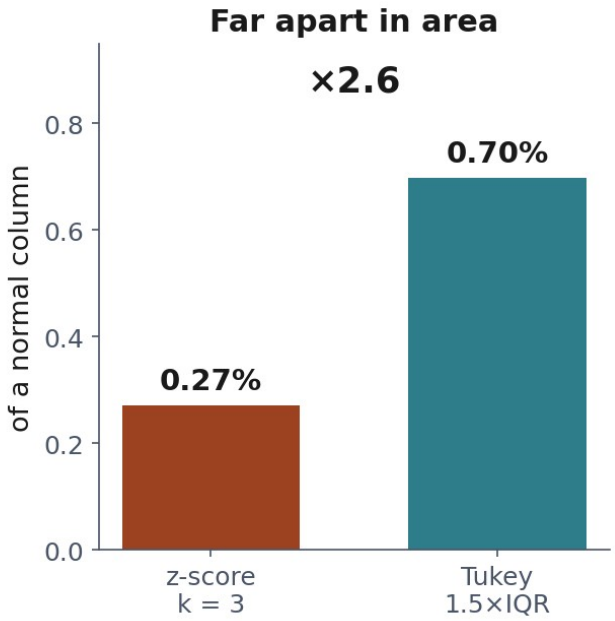
Rule 2: interquartile range (IQR)

$$[\ Q_1 - 1.5\ \text{IQR},\quad Q_3 + 1.5\ \text{IQR}\ ]$$

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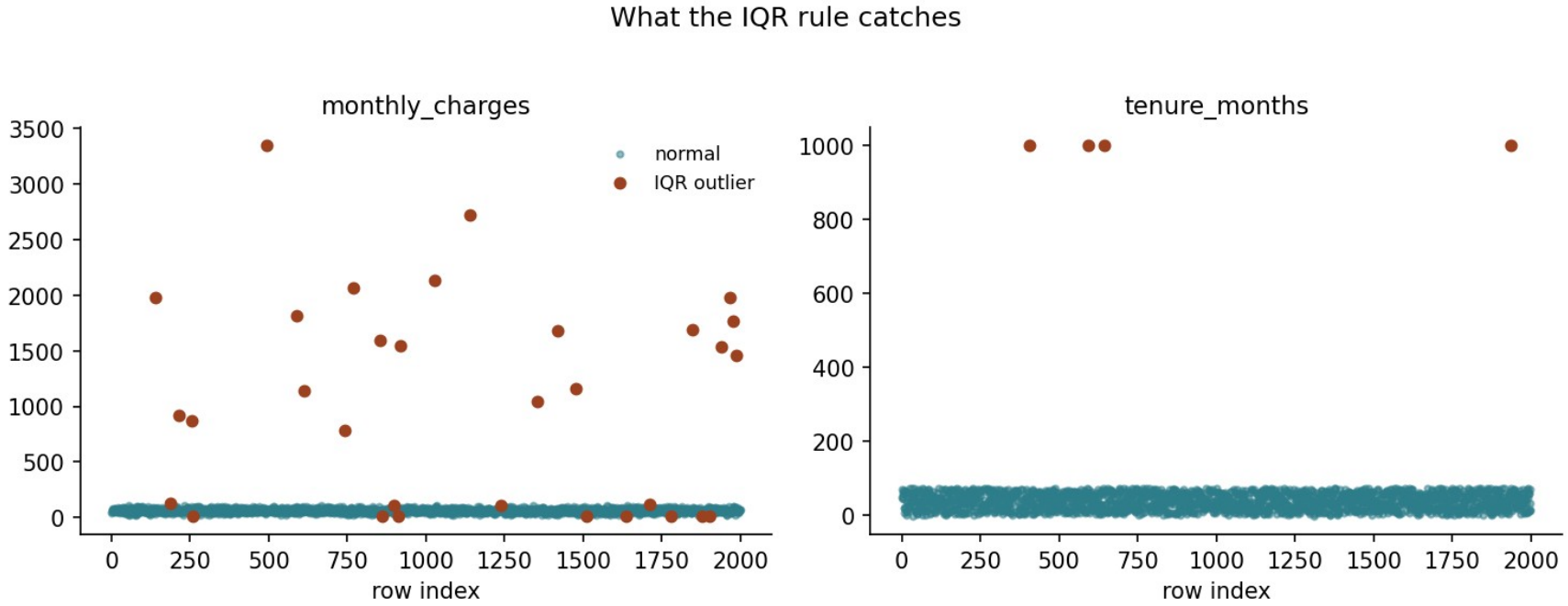
Close fences, very different counts





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Quartile rule: 12 false alarms



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What neither rule can tell you

`tenure_months` of **-3** is impossible. Neither rule flags it.

- -3 is not *extreme* relative to a column spanning 0–72
- It is not unusual, it is impossible: no tenure is negative. Checking validity needs a rule about the domain, not about the distribution

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Once something is flagged

The rules only find candidates. What to do with them is decided by someone who knows the **domain**.

- **Cap** it at the rule's upper limit: 3,344.7 becomes 110.0 (*winsorising*)
- **Remove** the row
- **Correct** it, if the true value is recoverable
- **Leave** it: unusual is not the same as wrong

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Notebook 1, live

section, and its slides	you call (pandas, no model yet)
first look, 6	`df.info()`, `df.describe()`
distributions, 7	`df.hist()`
missingness, 8-12	`df.isna().mean()`, and the mean before and after filling
outliers from scratch, 15-19	`.mean()` and `.std()` for rule 1, `.quantile([0.25, 0.75])` for rule 2
correlations, categories	`df.corr()`, `.value_counts()`

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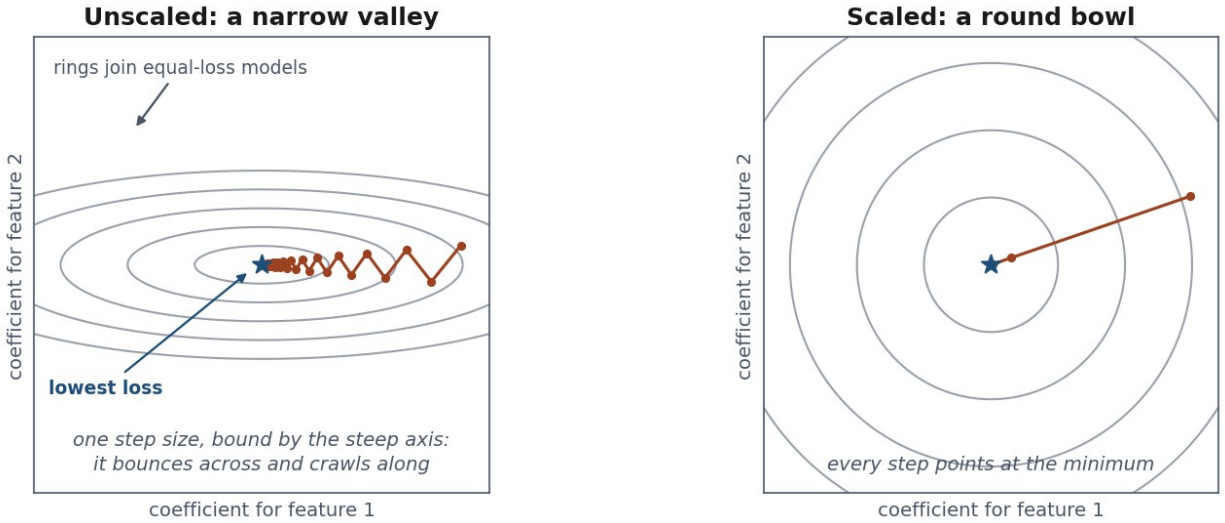
Break

Why scale?

- `tenure_months`: 0–72. `monthly_charges`: 15–128
- **Training**: choose w_1, w_2 to minimise Lesson 1's average loss, a paraboloid over (w_1, w_2). Lesson 1's slide 20 drew it for one number, the threshold
- **Gradient descent** repeats $w \leftarrow w - \alpha \nabla \hat{R}(w)$: a step downhill, its size set by the **learning rate** α, for every column. Can one α suit both?

Unequal scales: a narrow valley

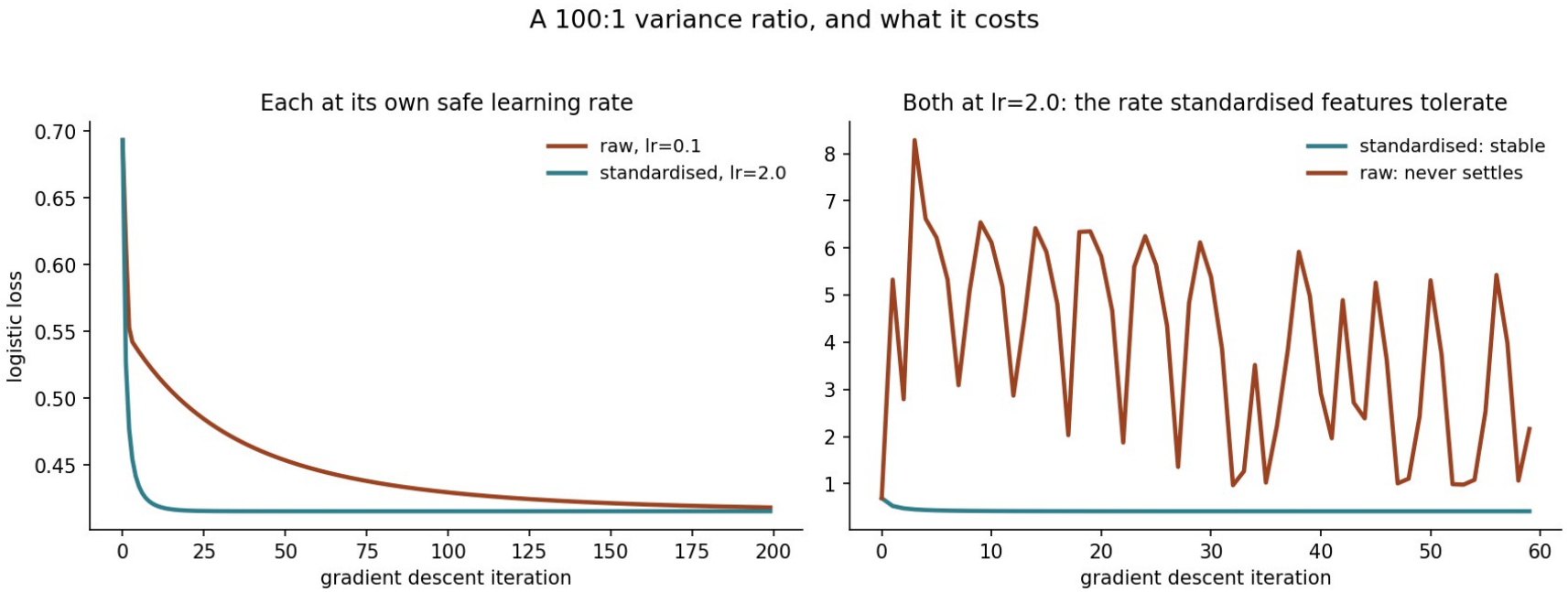
The average loss as a function of the two coefficients, seen from above



each rust dot is one step: $w \leftarrow w - \alpha \nabla \hat{R}(w)$, the same learning rate α in both directions

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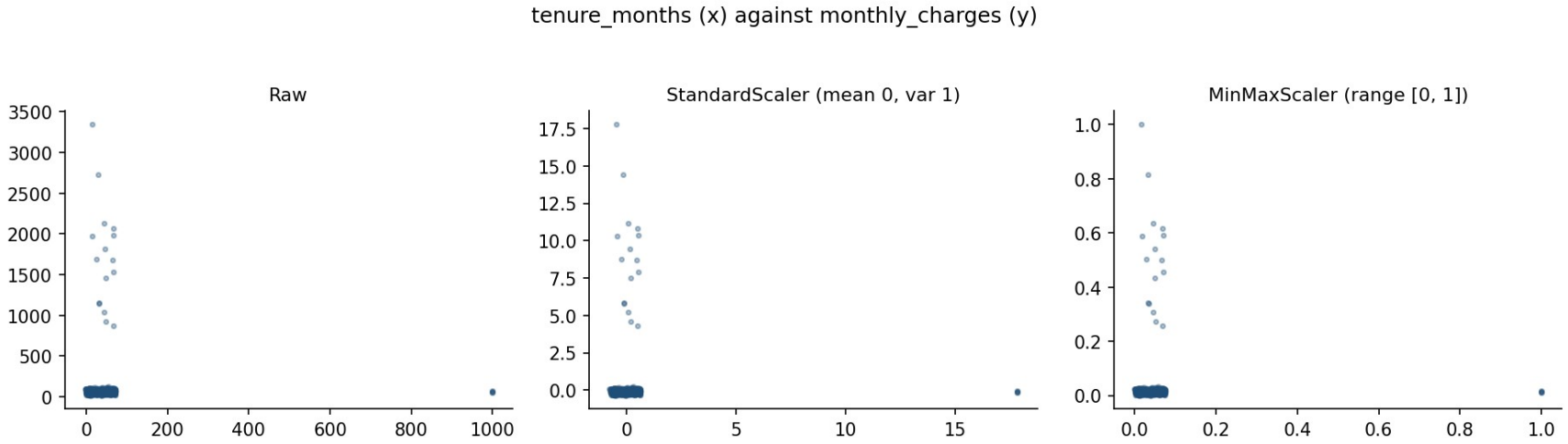
A 100:1 variance ratio, and what it costs



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StandardScaler vs MinMaxScaler

(x − μ)/σ and (x − min)/(max − min), both fitted on training data only.



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Every encoding makes a claim about the categories

A model needs numbers, and `"month-to-month"` is text: encoding turns it into numbers.

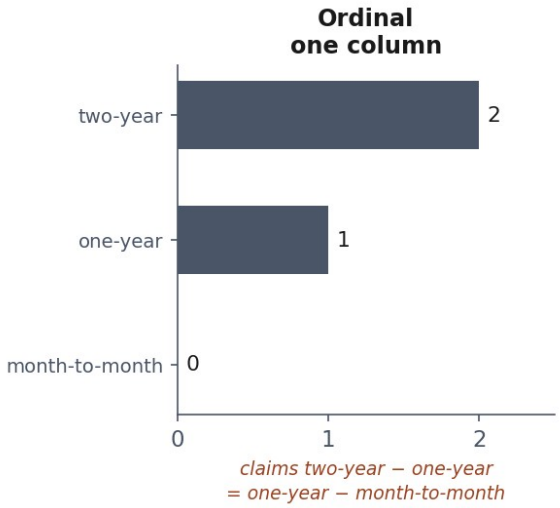
contract_type	rows	churn rate
month-to-month	1,092	0.266
one-year	488	0.137
two-year	420	0.071

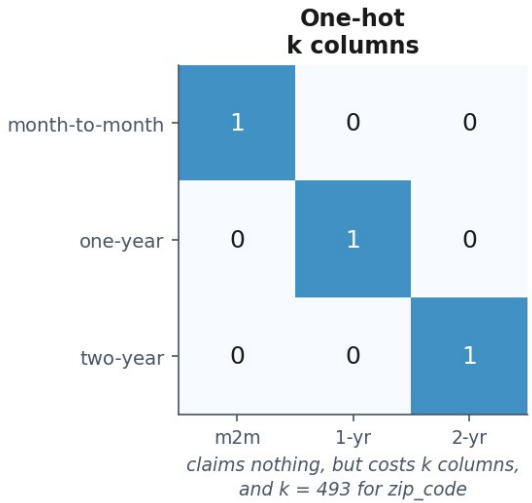
One customer, three encodings

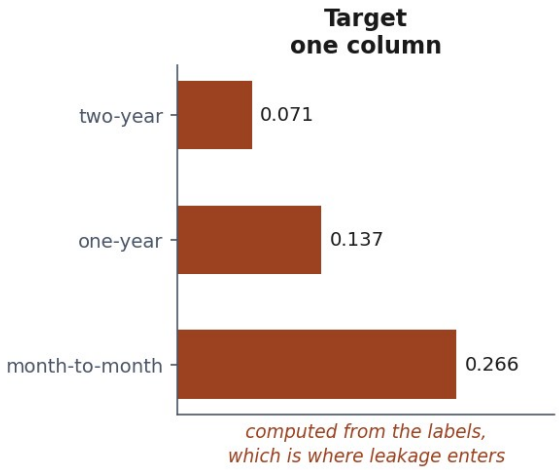
Encoding	A one-year customer becomes	What that asserts
One-hot	`[0, 1, 0]`	all different, none closer to another
Ordinal	`1`	ordered, **and** equally spaced
Target	`0.137`	replaced by the churn rate of its own category

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Ordinal: equal steps, unequal churn







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One-hot: no order assumed, but wide

- A column with k categories becomes k columns of 0s and 1s, a single 1 per row: `OneHotEncoder`
- No order between categories, all equally far apart: that is why it is the default
- The price is width: `zip_code` would add **493** columns to 2,000 rows
- Target encoding uses one column however many categories there are, but it is computed from the target - which is how leakage gets in

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One one-hot column too many

Most models include a constant column of ones, the **intercept**. The k one-hot columns (*dummies*) also always add up to 1:

$$\sum_{j=1}^{k} \text{dummy}_j = 1 = \text{intercept}$$

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4 columns, but only rank 3

Why all k dummies plus an intercept cannot work

contract_type	m2m	1y	2y	sum	intercept
month-to-month	**1**	0	0	**= 1**	**1**
one-year	0	**1**	0	**= 1**	**1**
two-year	0	0	**1**	**= 1**	**1**

the k dummy columns always sum to the intercept column: rank deficient by exactly one

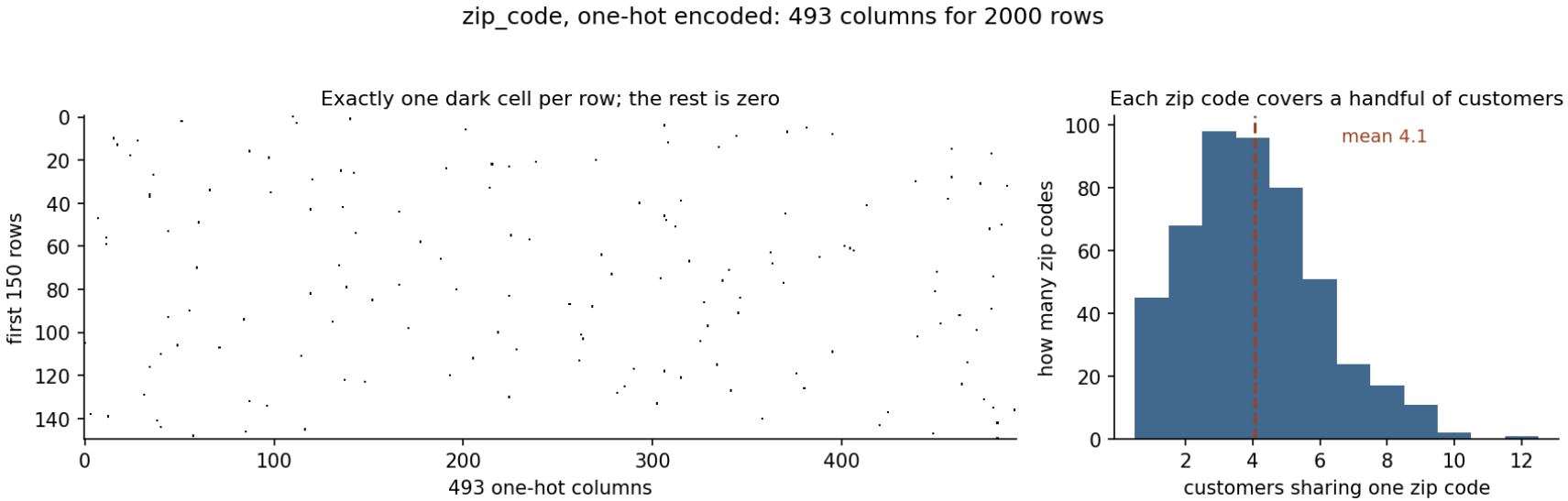
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The curse of dimensionality, briefly

One-hot `zip_code`: **493** new columns, mostly zero.

- Methods based on distances (Lesson 6): all points end up roughly equally far apart
- More coefficients to estimate from the same 2,000 rows: noisier estimates

zip_code: 493 codes, 2,000 customers



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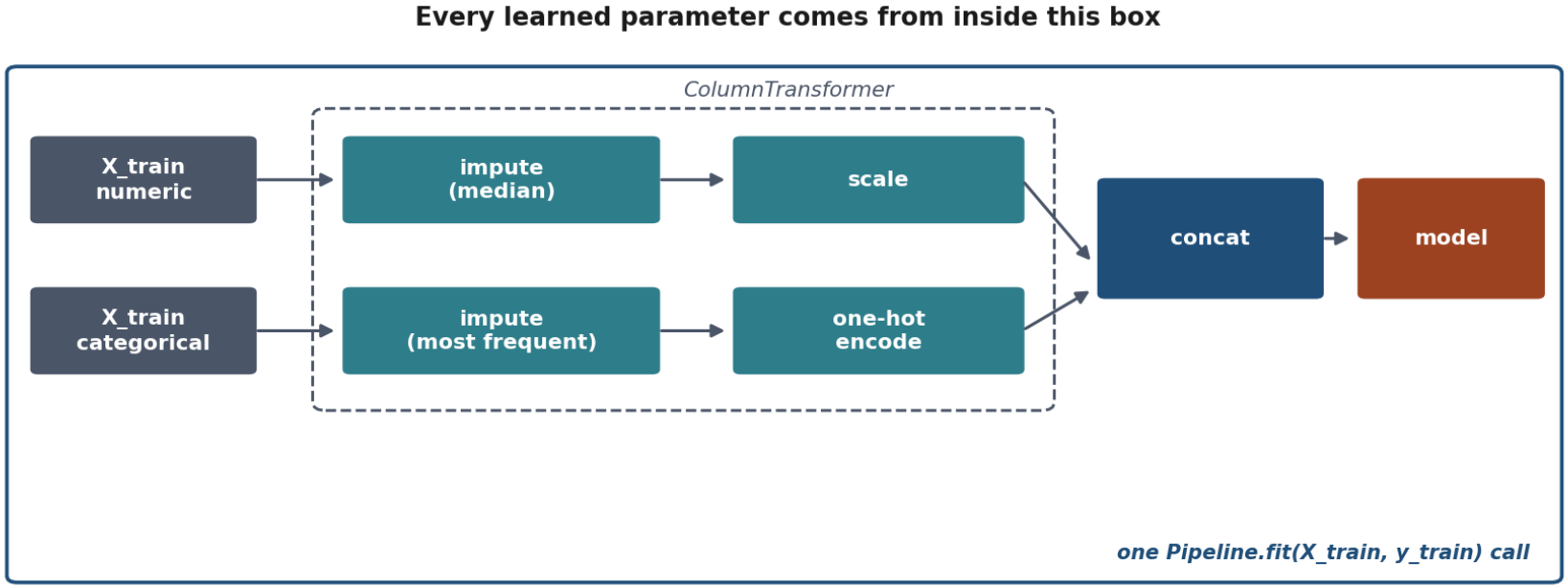
Preprocessing is part of f

f is the whole pipeline: **every** number learned from data - the scaler’s means, the imputer’s fill values - not only the model’s coefficients.

$$\mathrm{E}_{T \sim \mathcal{D}^m}\left[\hat{R}_T(f)\right] = R(f)$$

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ColumnTransformer + Pipeline



The code

```
prep = ColumnTransformer([
    ("num", num_pipe, num_cols),
    ("cat", cat_pipe, cat_cols),
])
model = Pipeline([
    ("prep", prep),
    ("clf", clf),
])
model.fit(X_train, y_train)
```

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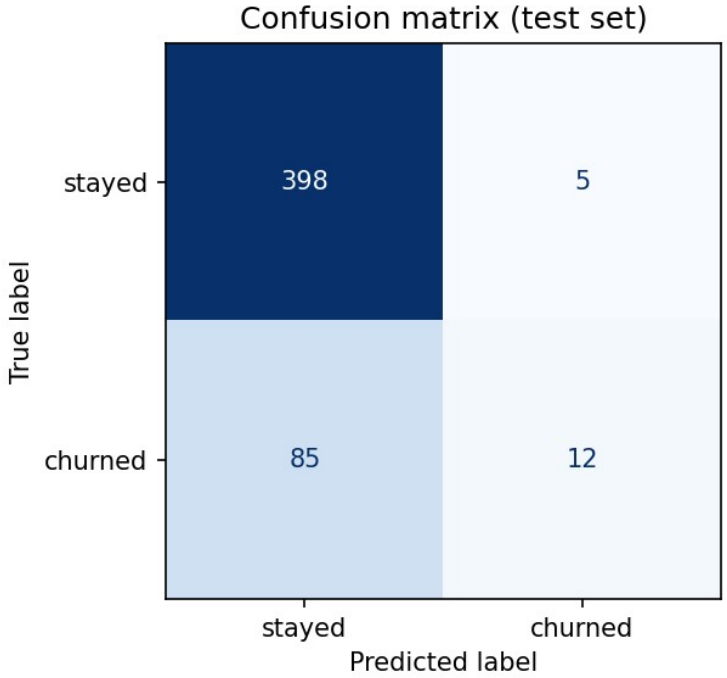
The model, on churn

- Accuracy: baseline **0.806**, model **0.820** (numeric + low-cardinality categoricals, no zip)
- Area under the receiver operating characteristic curve (AUC): **0.751**, the chance a random churner scores above a random non-churner; 0.5 is a coin flip
- In code: `LogisticRegression` against `DummyClassifier`, `roc_auc_score`

Accuracy barely beats the baseline. Why?

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Where the errors fall



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Feature engineering

New columns built from existing ones, which a linear model cannot build itself.

`total_paid` = charge × months stayed: AUC 0.7514 → 0.7548. Fix the 999s in `tenure_months` first and it scores **0.7439**: the gain came from the errors.

In code: `FunctionTransformer(add_total_paid)`, inside the pipeline

Notebook 2, live

section, and its slides	you call → the number to come back with
why scale, 24-27	gradient descent by hand → ratio **110:1** (built as 100:1); rate **2.0** scaled, **0.1** raw; raw at 2.0 swings **0.69 to 8.29**
encoding, 28-33	`pd.get_dummies`, `np.linalg.matrix_rank` → **4 columns, rank 3**
the pipeline, 36-38	`train_test_split`, then `ColumnTransformer` + `Pipeline`: `SimpleImputer`, `StandardScaler`, `OneHotEncoder`, `LogisticRegression`
the result, 39-40	`DummyClassifier`, `accuracy_score`, `roc_auc_score`, `ConfusionMatrixDisplay`
feature engineering, 41	`FunctionTransformer` → **0.7514 → 0.7548 → 0.7439**

Same rule, broken three ways

Same rule, broken three ways: only the first one looks like a bug

Lesson 1	select features on the full dataset *visible: an obvious extra step*
Today, leak 1	impute a missing value on the full dataset *invisible: looks like ordinary cleaning*
Today, leak 2	encode a category on the full dataset *invisible: two unremarkable lines of code*

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Leak 1: impute before splitting

`KNNImputer` fills a gap with the average of the five most similar rows: its **donors**.

Fit it on training and test rows together, and some donors are test rows: test information enters the training data.

The smoking gun

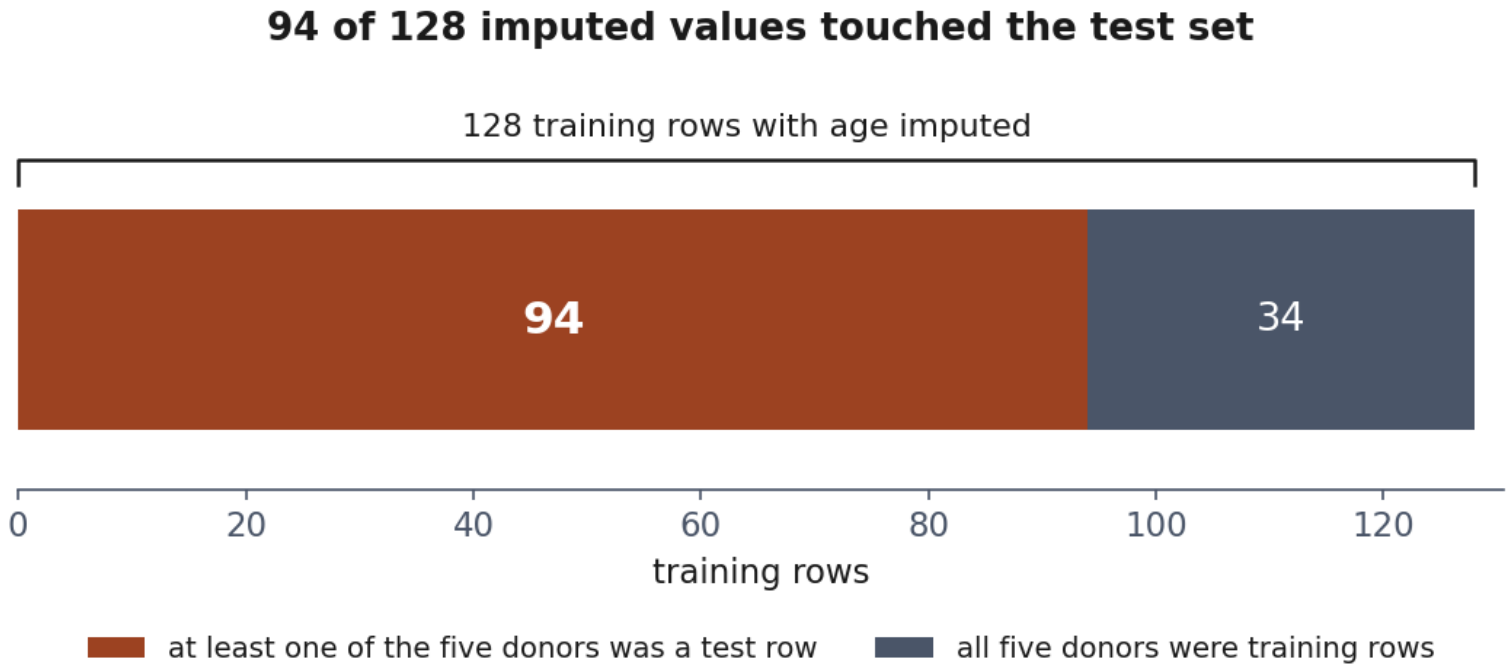
94 of 128 training rows with missing age had a **test-set row** among the five donors used to fill it.

In code: `nan_euclidean_distances`, the distance the imputer itself uses, finds the donors.

Nothing raised an error.

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Leakage, counted



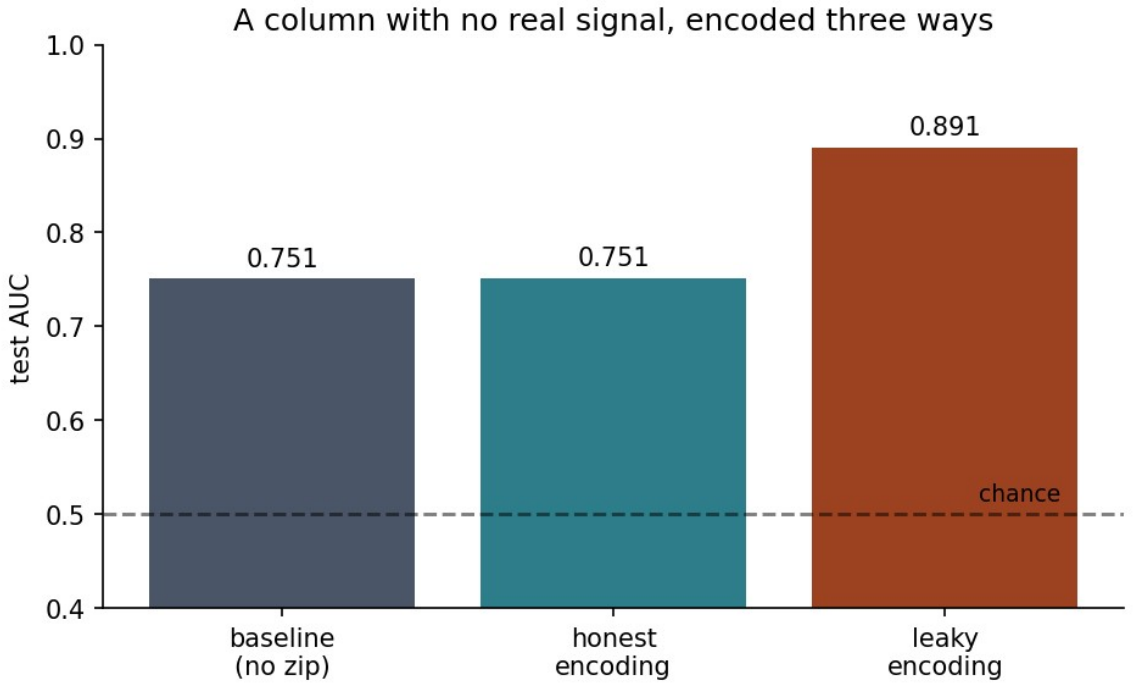
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Leak 2: encode before splitting

Replace `zip_code` with the **average churn rate of its own customers**, computed on all 2,000 customers, before the split.

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A column with no real signal, encoded three ways



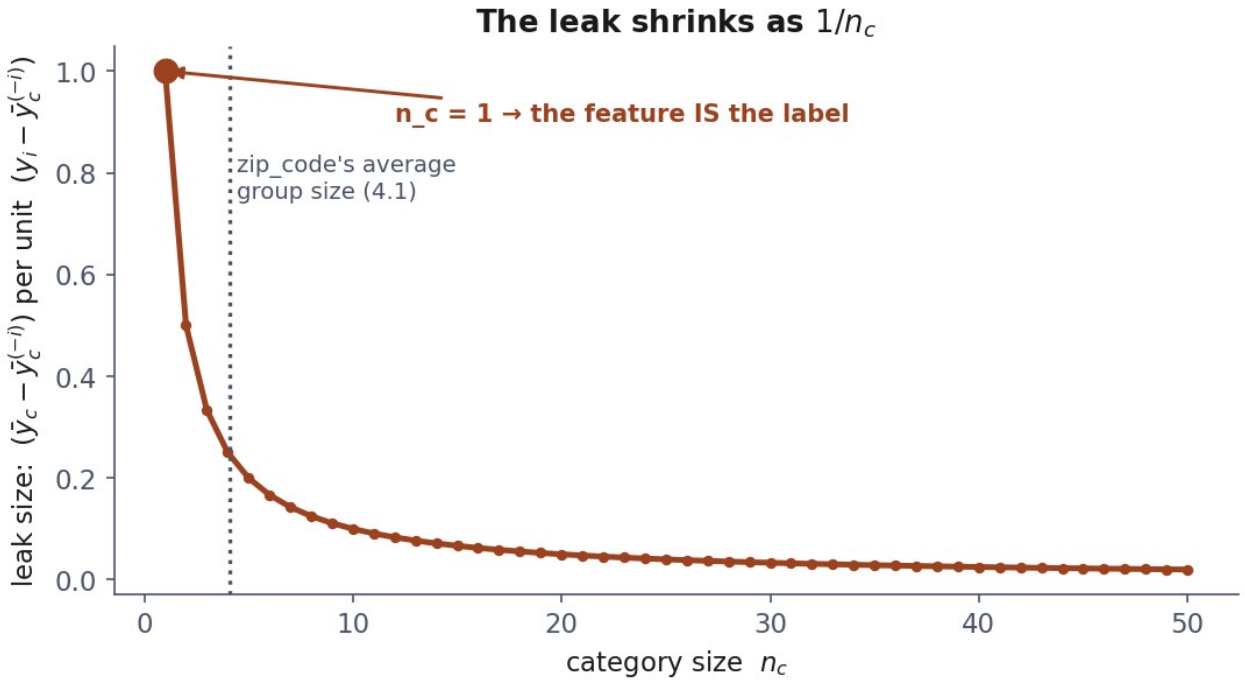
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How much of its own label a row sees

$$\bar{y}_c - \bar{y}_c^{(-i)} = \frac{y_i - \bar{y}_c^{(-i)}}{n_c}$$

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The leak shrinks as 1/n_c



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The fix

`sklearn.preprocessing.TargetEncoder`, inside the pipeline.

It splits the training rows into five parts and encodes each row from the **other four**: never from its own label.

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The rule, restated

Everything that **learns from data** is fitted on the training rows only, inside the pipeline.

For example: a mean, a median, the nearest neighbours, a churn rate per category, the edges of bins.

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Notebook 3, live

leak, and its slides	you call → what it shows
imputation, 44-46	`KNNImputer` fitted on every row; `nan_euclidean_distances`, its own metric, names the donors → **94 of 128**
encoding, 47-50	churn rate per `zip_code`, before `train_test_split` → AUC **0.891** from a column with no signal
the fix, 51	`TargetEncoder` inside the `ColumnTransformer` → the honest **0.751**
your own split	`clone` both models, refit on a split of your own
scored with	`roc_auc_score` throughout

What we did today

- Explored before transforming: types, gaps, shape, correlations
- Diagnosed *why* values are missing, not just *how many*
- Two outlier rules, and what neither can tell you
- Scaling, encoding, engineering: all inside `Pipeline`
- Two leaks that raise no error and look like good results

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Homework: we discuss it on Friday 16 October

`Exercises/02_data_exploration_and_preparation.md`

Build the full preparation pipeline yourself, and **find a leak on purpose** before fixing it.

Before next week

- Work the three notebooks in order
- Read the handout: it has the derivations behind today’s slides
- Take the quiz
- **Do the exercise**

Next: regression, the first model we derive completely.

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